Physics 9th – Chapter-3-Dynamics-I (Short Response Questions)

Q1. When a motor cyclist hit a stationary car, he may fly off the motor cycle and driver in the car may get neck injury. Explain

Ans:

- According to Newton's first law (inertia), on collision the motorcycle stops suddenly but the cyclist's body continues its state of motion and flies off.
- In the car, the lower body moves with the car as it is pushed by the collision, while the head and upper body tend to stay at rest due to inertia.
- This sudden relative motion jerks the neck and causes neck injury.

Q2. In autumn, when you shake a branch, the leaves are detached. Why?

Ans:

- Due to inertia of rest (Newton's first law), the leaves tend to remain at rest.
- When the branch is shaken suddenly, it moves but the leaves, which are loosely attached, resist this change in motion.
- They separate from the branch and fall down under gravity.

Q3. Why it is not safe to apply brakes only on the front wheel of a bicycle?

Ans:

- Braking the front wheel stops it suddenly, but due to inertia of motion the rear wheel and the rider's body tend to continue moving forward.
- The rear wheel lifts up and the rider is thrown forward, causing an accident.
- Therefore both brakes should be applied gradually.

Q4. Deduce Newton's first law of motion from Newton's second law of motion.

Ans:

- According to Newton's second law:

$$F = ma$$

- Acceleration is the rate of change of velocity, $a = (v_f - v_i)/t$. Putting this in the equation:

$$F = m\,(v_f - v_i) / t$$

- If no external force acts on the body, $F = 0$:

$$m\,(v_f - v_i) / t = 0$$

- Since mass $m \neq 0$ and time $t \neq 0$, the velocity difference must be zero:

$$v_f - v_i = 0$$

$$v_f = v_i$$

- Conclusion: The velocity does not change, i.e. $a = 0$. So a body at rest ($v_i = 0$) remains at rest ($v_f = 0$), and a body in motion continues with uniform velocity v_i, unless an external force acts on it. This is Newton's first law.

Q5. Action and reaction are equal but opposite in direction. These forces always act in pair. Do they balance each other? Can bodies move under action - reaction pair?

Ans:

- Balanced forces act on the same body, so they cancel out.
- Action and reaction forces act on different bodies, so they do not cancel out or balance each other.
- Yes, bodies can move under an action-reaction pair. Example: a rocket pushes the gases downward (action), and the gases exert an equal and opposite upward force on the rocket (reaction), which propels the rocket upward.

Q6. A man slips on the oily floor; he wants to move out of this area. He is alone. He throws his bag to move out of this slippery area. Why is it so?

Ans:

- By Newton's third law, when he throws the bag in one direction (action), the bag exerts an equal and opposite force on him (reaction).
- Since friction on the oily floor is minimum, this reaction force moves him in the opposite direction, out of the slippery area.

Q7. How would you use Newton's 3rd law of motion and law of conservation of momentum to explain motion of rocket?

Ans:

- According to Newton's third law, the rocket exerts an action force on the burning gases to expel them backward, and the gases exert an equal and opposite reaction force on the rocket, pushing it forward.
- By the law of conservation of momentum, the backward momentum of the gases equals the forward momentum gained by the rocket, so the total change in momentum is zero.

Q8. Why are cricket batter gloves padded with foam?

Ans:

- The rate of change of momentum gives force:

$$F = \Delta p / \Delta t$$

$$F \propto 1 / \Delta t$$

- Foam increases the impact time (Δt) for the same change in momentum (Δp), so the force on the batter's hand is reduced and the hand is protected.

Q9. Where will your weight be greater, near earth or near moon? What about mass?

Ans:

- Weight ($W = mg$) is greater near Earth because the gravitational acceleration of Earth is greater than that of the Moon (g on Moon is about 1/6 of Earth).
- Mass is the amount of matter in a body, so it remains the same at both places.

Q10. When Ronaldo kicks the ball, at the highest point of balls both Earth and ball attracts each other with the same magnitude of force. Why then the ball moves towards Earth and not the Earth?

Ans:

- By Newton's third law, the force between Earth and ball is equal.
- But acceleration is a = F/m, so it is inversely proportional to mass.
- Earth's mass is enormous, so its acceleration is negligible, while the ball's small mass gives it a noticeable acceleration towards Earth.